Q .1

QUESTIONS

Q1 Student Management System [BL3 – Apply] 10 marks
Q2 Library Book Management [BL3 – Apply] 10 marks
Q3 Employee Salary Calculation [BL3 – Apply] 10 marks
Q4 Shape Area Calculator using Polymorphism [BL3 – Apply] 10 marks
Q5 5. Vehicle Registration System [BL3 – Apply] 10 marks
Q6 6. Bank Account Hierarchy [BL4 – Analyze] 10 marks
Q7 7. Hospital Patient Record System [BL4 – Analyze] 10 marks

Q1 Student Management System [BL3 – Apply] 10 marks

Write pseudocode to design a Student class with attributes (name, rollNo, marks) and methods to:
- Accept student details
- Calculate average marks
- Display result (Pass/Fail)

Apply encapsulation and data hiding by restricting direct access to attributes.
OOP Concepts: Encapsulation | Data Hiding | Classes & Objects

Your Answer

```
START

CLASS student

      PRIVATE
          name
          rollNo
          marks1,marks2,marks3
          average

       PUBLIC

       METHOD acceptDetails()
              INPUT name
              INPUT rollNo
              INPUT  marks1
              INPUT  marks2
              INPUT  marks3
       END METHOD
```

– Apply] 10 marks
Q3 Employee Salary Calculation [BL3 – Apply] 10 marks
Q4 Shape Area Calculator using Polymorphism [BL3 – Apply] 10 marks
Q5 5. Vehicle Registration System [BL3 – Apply] 10 marks
Q6 6. Bank Account Hierarchy [BL4 – Analyze] 10 marks
Q7 7. Hospital Patient Record System [BL4 – Analyze] 10 marks
Q8 8. E-Commerce Product Catalog [BL4 – Analyze] 10 marks
Q9 9. University Staff Hierarchy [BL4 – Analyze] 10 marks

Apply encapsulation and data hiding by restricting direct access to attributes.
OOP Concepts: Encapsulation | Data Hiding | Classes & Objects

Your Answer

```
       METHOD  calculateAverage()
              average<-(marks1+marks2+marks3)/3
       END METHOD

       METHOD displayResult()
             DISPLAY "Name =",name
             DISPLAY "Roll no=",rollNo
             DISPLAY "Average=",average

             IF average>=35 THEN
               DISPLAY "Result = PASS"
             ELSE
               DISPLAY "Result=FAIL"
             END IF
        END METHOD

END CLASS

CREATE object s OF Student

CALL s.acceptDetails()
CALL s.calculateAverage()
CALL s.displayResult()

STOP
```

Save Answer

Q.2

[BL3 – Apply] 10 marks
Q4 Shape Area Calculator using Polymorphism [BL3 – Apply] 10 marks
Q5 5. Vehicle Registration System [BL3 – Apply] 10 marks
Q6 6. Bank Account Hierarchy [BL4 – Analyze] 10 marks
Q7 7. Hospital Patient Record System [BL4 – Analyze] 10 marks
Q8 8. E-Commerce Product Catalog [BL4 – Analyze] 10 marks
Q9 9. University Staff Hierarchy [BL4 – Analyze] 10 marks
Q10 10. Smart Home Device Controller [BL4 – Analyze] 10 marks

```
    METHOD issueBook()
        IF isAvailable =TRUE THEN
             isAvailable <-FALSE
             DISPLAY "Book Issued"
        ELSE
            DISPLAY "Book Not Available"
        END IF
    END METHOD

    METHOD returnBook()
        IF isAvailable =FALSE THEN
            isAvailable <-TRUE
            DISPLAY "Book Returned"
        ELSE
            DISPLAY "Book Already Available"
        END IF
    END METHOD

    METHOD displayBook()
           DISPLAY "Title : ",title
           DISPLAY "Author :",author
           DISPLAY "ISBN: ",ISBN
           DISPLAY "Available :",isAvailable
     END METHOD

END CLASS

CREATE  Book B1
CREATE  Book B2
CREATE  Book B3
```

– Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

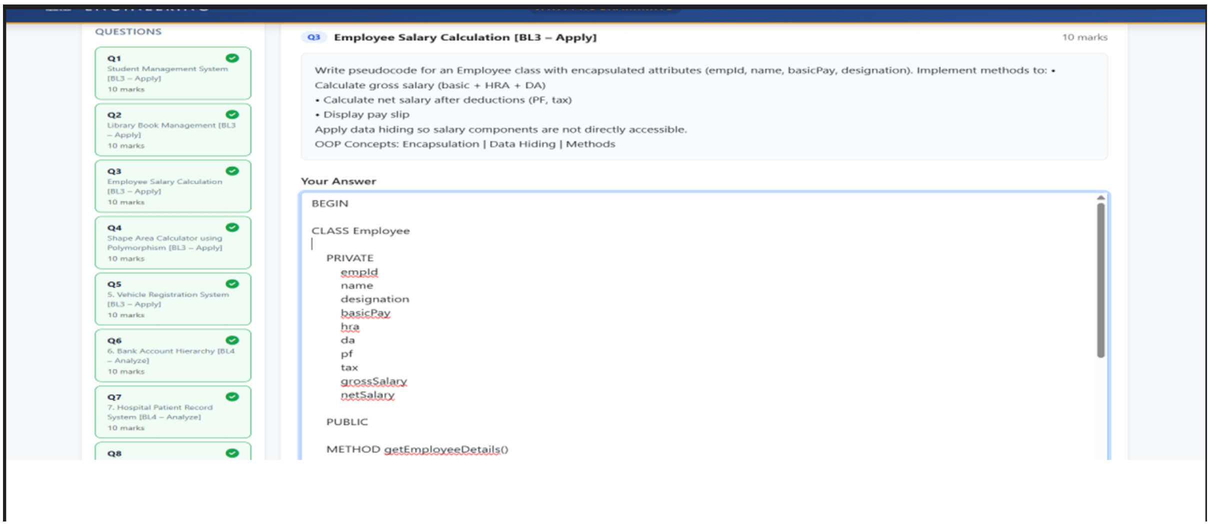
Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

```
            DISPLAY "Author :",author
            DISPLAY "ISBN: ",ISBN
            DISPLAY "Available :",isAvailable
        END METHOD

END CLASS

CREATE  Book B1
CREATE  Book B2
CREATE  Book B3

CALL  B1.setBookDetails("java","james",101)
CALL  B2.setBookDetails("python","sri",201)
CALL  B3.setBookDetails("C++","baji",301)

CALL  B1.issueBook()
CALL  B2.issueBook()
CALL  B3.issueBook()

CALL  B1.displayBook()
CALL  B2.displayBook()
CALL  B3.displayBook()

STOP
```

Save Answer

Q.3



QUESTIONS

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

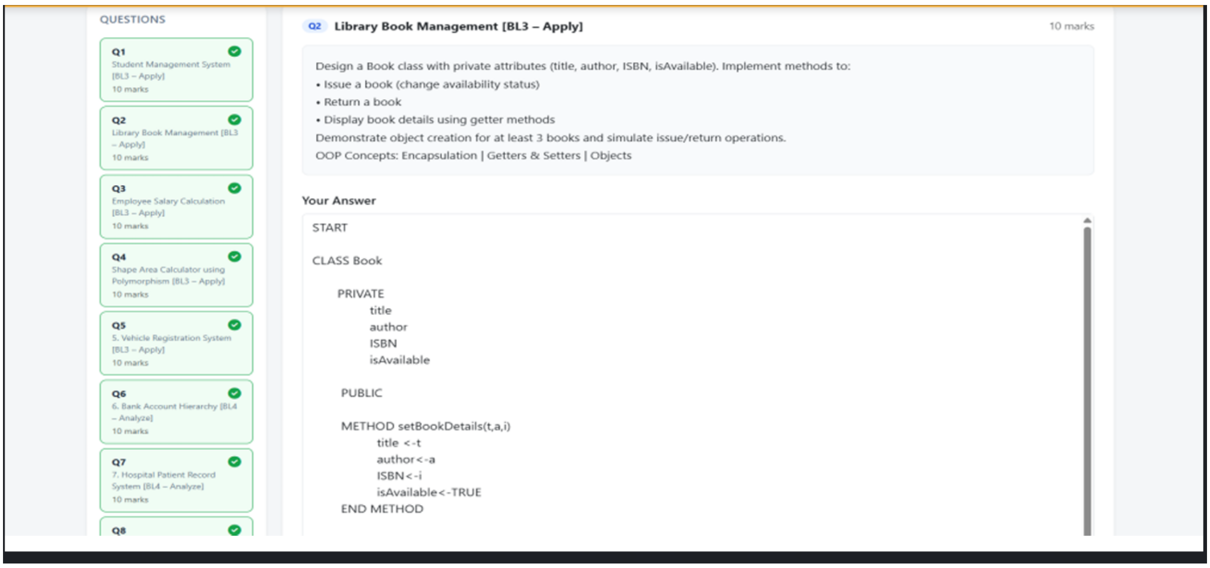
Q8

Q3 **Employee Salary Calculation [BL3 – Apply]** 10 marks

Write pseudocode for an Employee class with encapsulated attributes (empId, name, basicPay, designation). Implement methods to: • Calculate gross salary (basic + HRA + DA)
• Calculate net salary after deductions (PF, tax)
• Display pay slip
Apply data hiding so salary components are not directly accessible.
OOP Concepts: Encapsulation | Data Hiding | Methods

Your Answer

```
BEGIN

CLASS Employee

    PRIVATE
        empId
        name
        designation
        basicPay
        hra
        da
        pf
        tax
        grossSalary
        netSalary

    PUBLIC

    METHOD getEmployeeDetails()
```



QUESTIONS

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8

Q2 **Library Book Management [BL3 – Apply]** 10 marks

Design a Book class with private attributes (title, author, ISBN, isAvailable). Implement methods to:
• Issue a book (change availability status)
• Return a book
• Display book details using getter methods
Demonstrate object creation for at least 3 books and simulate issue/return operations.
OOP Concepts: Encapsulation | Getters & Setters | Objects

Your Answer

```
START

CLASS Book

    PRIVATE
            title
            author
            ISBN
            isAvailable

    PUBLIC

    METHOD setBookDetails(t,a,i)
            title <-t
            author<-a
            ISBN<-i
            isAvailable<-TRUE
    END METHOD
```

[BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

```
METHOD getEmployeeDetails()

   INPUT empId
   INPUT name
   INPUT designation
   INPUT basicPay

END METHOD


METHOD calculateGrossSalary()

   hra ← basicPay × 20 / 100
   da ← basicPay × 10 / 100

   grossSalary ← basicPay + hra + da

END METHOD


METHOD calculateNetSalary()

   pf ← basicPay × 12 / 100

   tax ← grossSalary × 5 / 100

   netSalary ← grossSalary − (pf + tax)

END METHOD


METHOD displayPaySlip()
```

[BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

```
   METHOD displayPaySlip()

      PRINT "----------- PAY SLIP -----------"

      PRINT "Employee ID      :", empId
      PRINT "Employee Name    :", name
      PRINT "Designation      :", designation

      PRINT "Basic Pay        :", basicPay
      PRINT "HRA              :", hra
      PRINT "DA               :", da

      PRINT "Gross Salary     :", grossSalary

      PRINT "PF Deduction     :", pf
      PRINT "Tax Deduction    :", tax

      PRINT "Net Salary       :", netSalary

      PRINT "-------------------------------"

   END METHOD

END CLASS


DECLARE emp AS Employee

CALL emp.getEmployeeDetails()

CALL emp.calculateGrossSalary()
```

[BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

```
      PRINT "Gross Salary     :", grossSalary

      PRINT "PF Deduction     :", pf
      PRINT "Tax Deduction    :", tax

      PRINT "Net Salary       :", netSalary

      PRINT "-------------------------------"

   END METHOD

END CLASS


DECLARE emp AS Employee

CALL emp.getEmployeeDetails()

CALL emp.calculateGrossSalary()

CALL emp.calculateNetSalary()

CALL emp.displayPaySlip()

END
```

Save Answer

Q.4

QUESTIONS

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8

Q4 Shape Area Calculator using Polymorphism [BL3 – Apply]

Create a Shape base class with an overridable calculateArea() method. Apply this to:
- Circle — area using radius
- Rectangle — area using length and breadth
- Triangle — area using base and height

Use method overriding to demonstrate runtime polymorphism. Call calculateArea() for each object.
OOP Concepts: Inheritance | Polymorphism | Method Overriding

Your Answer

```
BEGIN

CLASS Shape
   METHOD calculateArea()
      DISPLAY "Area Calculation"
   END METHOD
END CLASS

CLASS Circle EXTENDS Shape
   DECLARE radius

   CONSTRUCTOR(radius)
      SET this.radius = radius
   END CONSTRUCTOR

   METHOD calculateArea()
      area = 3.14 × radius × radius
      DISPLAY "Area of Circle = ", area
   END METHOD
```

Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

Your Answer

```
END CLASS

CLASS Rectangle EXTENDS Shape
   DECLARE length, breadth

   CONSTRUCTOR(length, breadth)
      SET this.length = length
      SET this.breadth = breadth
   END CONSTRUCTOR

   METHOD calculateArea()
      area = length × breadth
      DISPLAY "Area of Rectangle = ", area
   END METHOD
END CLASS

CLASS Triangle EXTENDS Shape
   DECLARE base, height

   CONSTRUCTOR(base, height)
      SET this.base = base
      SET this.height = height
   END CONSTRUCTOR

   METHOD calculateArea()
      area = (base × height) / 2
      DISPLAY "Area of Triangle = ", area
   END METHOD
END CLASS

CREATE Shape reference s
```

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

```
    DECLARE base, height

    CONSTRUCTOR(base, height)
        SET this.base = base
        SET this.height = height
    END CONSTRUCTOR

    METHOD calculateArea()
        area = (base × height) / 2
        DISPLAY "Area of Triangle = ", area
    END METHOD
END CLASS

CREATE Shape reference s

SET s = new Circle(7)
CALL s.calculateArea()

SET s = new Rectangle(10, 5)
CALL s.calculateArea()

SET s = new Triangle(8, 6)
CALL s.calculateArea()

END
```

Save Answer

Q.5

QUESTIONS

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8

Q5 5. Vehicle Registration System [BL3 – Apply]

Design a Vehicle class with attributes (regNo, owner, fuelType). Extend it into:

- TwoWheeler — with engine capacity
- FourWheeler — with seating capacity and AC status

Apply inheritance to reuse common attributes and override a displayDetails() method in each subclass.

OOP Concepts: Inheritance | Method Overriding | Subclass

Your Answer

```
BEGIN

CLASS Vehicle

    DECLARE regNo
    DECLARE owner
    DECLARE fuelType

    METHOD displayDetails()
        DISPLAY "Registration Number :", regNo
        DISPLAY "Owner :", owner
        DISPLAY "Fuel Type :", fuelType
    END METHOD

END CLASS


CLASS TwoWheeler INHERITS Vehicle

    DECLARE engineCapacity
```

Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

```
    DECLARE engineCapacity

    METHOD displayDetails()

        DISPLAY "----- Two Wheeler Details -----"
        DISPLAY "Registration Number :", regNo
        DISPLAY "Owner :", owner
        DISPLAY "Fuel Type :", fuelType
        DISPLAY "Engine Capacity :", engineCapacity

    END METHOD

END CLASS


CLASS FourWheeler INHERITS Vehicle

    DECLARE seatingCapacity
    DECLARE acStatus

    METHOD displayDetails()

        DISPLAY "----- Four Wheeler Details -----"
        DISPLAY "Registration Number :", regNo
        DISPLAY "Owner :", owner
        DISPLAY "Fuel Type :", fuelType
        DISPLAY "Seating Capacity :", seatingCapacity
        DISPLAY "AC Status :", acStatus

    END METHOD

END CLASS
```

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

```
    END METHOD

END CLASS


CREATE OBJECT bike OF TwoWheeler

SET bike.regNo ← "AP39AB1234"
SET bike.owner ← "Srikanth"
SET bike.fuelType ← "Petrol"
SET bike.engineCapacity ← 150

CALL bike.displayDetails()


CREATE OBJECT car OF FourWheeler

SET car.regNo ← "TS09CD5678"
SET car.owner ← "Ravi"
SET car.fuelType ← "Diesel"
SET car.seatingCapacity ← 5
SET car.acStatus ← "Available"

CALL car.displayDetails()

END
```

Save Answer

Q.6

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8

Q6 **6. Bank Account Hierarchy [BL4 – Analyze]**

Develop pseudocode for a BankAccount superclass and two subclasses — SavingsAccount (with interest calculation) and CurrentAccount (with overdraft facility). Analyze and implement:

- Inheritance of common attributes (accNo, balance, holder)
- Method overriding for withdrawal behavior
- How overdraft protection differs from savings limit

Compare the behavior of withdraw() across both subclasses and justify the design choices.

OOP Concepts: Inheritance | Method Overriding | Polymorphism

Your Answer

```
BEGIN

CLASS BankAccount

    VARIABLE accountNumber
    VARIABLE holderName
    VARIABLE balance

    METHOD withdraw(amount)

        DISPLAY "Withdrawal"

    END METHOD

END CLASS


CLASS SavingsAccount EXTENDS BankAccount
```

ENGINEERING JAVA PROGRAMMING

10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

```
    VARIABLE interestRate

    METHOD withdraw(amount)

        IF amount <= balance THEN

            balance ← balance − amount

            DISPLAY "Withdrawal Successful"

        ELSE

            DISPLAY "Insufficient Balance"

        ENDIF

    END METHOD

END CLASS


CLASS CurrentAccount EXTENDS BankAccount

    VARIABLE overdraftLimit

    METHOD withdraw(amount)

        IF amount <= balance + overdraftLimit THEN

            balance ← balance − amount
```

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

```
        balance ← balance − amount

        DISPLAY "Withdrawal Successful"

      ELSE

        DISPLAY "Overdraft Limit Exceeded"

      ENDIF

   END METHOD

END CLASS

CREATE SavingsAccount

CREATE CurrentAccount

CALL SavingsAccount.withdraw(500)

CALL CurrentAccount.withdraw(1500)

DISPLAY SavingsAccount.balance

DISPLAY CurrentAccount.balance

END
```

Save Answer

Q.7

QUESTIONS

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q7 **7. Hospital Patient Record System [BL4 – Analyze]** 10 marks

Design a Patient base class and analyze how it extends into:
- InPatient — with ward number, admission date, and daily charges
- OutPatient — with appointment date and consultation fee

Analyze encapsulation requirements for sensitive data (diagnosis, prescription). Override a generateBill() method for each type and compare the billing logic differences. OOP Concepts: Encapsulation | Inheritance | Polymorphism

Your Answer

```
BEGIN

CLASS Patient

   PRIVATE diagnosis
   PRIVATE prescription

   VARIABLE patientName

   METHOD generateBill()

      RETURN 0

   END METHOD

END CLASS


CLASS InPatient EXTENDS Patient
```

5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

```
        RETURN dailyCharge × numberOfDays

    END METHOD

END CLASS


CLASS OutPatient EXTENDS Patient

    VARIABLE consultationFee

    METHOD generateBill()

        RETURN consultationFee

    END METHOD

END CLASS


CREATE InPatient

CREATE OutPatient

DISPLAY InPatient.generateBill()

DISPLAY OutPatient.generateBill()

END
```

Save Answer

Q.8

QUESTIONS

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8

Q8 **8. E-Commerce Product Catalog [BL4 – Analyze]** 10 marks

Analyze and design a Product class hierarchy for an e-commerce system:
• Electronics — with warranty, brand, voltage
• Clothing — with size, fabric, gender
• Grocery — with expiryDate and weight
Override an applyDiscount() method for each category with different discount rules. Analyze why polymorphism makes the cart checkout process flexible and extensible. OOP Concepts: Polymorphism | Inheritance | Method Overriding

Your Answer

```
BEGIN

CLASS Product

    VARIABLE productName
    VARIABLE price

    METHOD applyDiscount()
        RETURN price
    END METHOD

END CLASS


CLASS Electronics EXTENDS Product

    METHOD applyDiscount()

        RETURN price – (price × 0.10)
```

10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

```
  END METHOD

END CLASS

CLASS Clothing EXTENDS Product

  METHOD applyDiscount()

    RETURN price – (price × 0.20)

  END METHOD

END CLASS

CLASS Grocery EXTENDS Product

  METHOD applyDiscount()

    RETURN price – (price × 0.05)

  END METHOD

END CLASS

CREATE Electronics

CREATE Clothing
```

10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

CLASS Grocery EXTENDS Product

METHOD applyDiscount()

RETURN price – (price × 0.05)

END METHOD

END CLASS

CREATE Electronics

CREATE Clothing


```
CREATE Grocery

DISPLAY Electronics.applyDiscount()

DISPLAY Clothing.applyDiscount()

DISPLAY Grocery.applyDiscount()

END
```

Save Answer

Q.9

QUESTIONS

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8

Q9 **9. University Staff Hierarchy [BL4 – Analyze]** 10 marks

Analyze the design of a Staff base class extended by TeachingStaff and NonTeachingStaff. Further extend TeachingStaff into Professor and Lecturer (multilevel inheritance). For each level:
• Identify which attributes should be private vs protected
• Override calculateAllowance() at each level
• Analyze how protected access enables inheritance without full exposure Justify the use of multilevel inheritance and explain how access modifiers enforce data hiding.
OOP Concepts: Multilevel Inheritance | Access Modifiers | Data Hiding

Your Answer

```
BEGIN

CLASS Staff

   PRIVATE staffID
   PRIVATE name
   PROTECTED basicSalary

   METHOD calculateAllowance()
      RETURN 0
   END METHOD

END CLASS


CLASS TeachingStaff EXTENDS Staff

   METHOD calculateAllowance()
```

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

```
      RETURN basicSalary × 0.20
   END METHOD

END CLASS


CLASS Professor EXTENDS TeachingStaff

   METHOD calculateAllowance()
      RETURN basicSalary × 0.35
   END METHOD

END CLASS


CLASS Lecturer EXTENDS TeachingStaff

   METHOD calculateAllowance()
      RETURN basicSalary × 0.25
   END METHOD

END CLASS


CLASS NonTeachingStaff EXTENDS Staff

   METHOD calculateAllowance()
      RETURN basicSalary × 0.15
   END METHOD

END CLASS
```

10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

```
END CLASS

CLASS NonTeachingStaff EXTENDS Staff

   METHOD calculateAllowance()
      RETURN basicSalary × 0.15
   END METHOD

END CLASS

CREATE Professor
CREATE Lecturer
CREATE NonTeachingStaff

DISPLAY Professor.calculateAllowance()

DISPLAY Lecturer.calculateAllowance()

DISPLAY NonTeachingStaff.calculateAllowance()

END
```

Save Answer

Q.10

QUESTIONS

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8

Q10 **10. Smart Home Device Controller [BL4 – Analyze]** 10 marks

Analyze and design a SmartDevice superclass with subclasses: SmartLight, SmartThermostat, and SmartLock. Override an executeCommand(cmd) method in each, where the same command (e.g., 'on', 'off', 'status') produces device-specific behavior. Analyze:
• How polymorphism allows a single controller loop to manage all devices
• Security implications of data hiding in SmartLock (PIN must never be exposed) Design the class hierarchy and explain the OOP principles applied at each level.
OOP Concepts: Polymorphism | Encapsulation | Inheritance

Your Answer

```
BEGIN

CLASS SmartDevice
   VARIABLE deviceName

   CONSTRUCTOR(name)
      deviceName ← name
   END CONSTRUCTOR

   METHOD executeCommand(command)
      DISPLAY "Generic Device"
   END METHOD
END CLASS


CLASS SmartLight EXTENDS SmartDevice

   METHOD executeCommand(command)
```

10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

```
    IF command = "on" THEN
      DISPLAY "Light Switched ON"

    ELSE IF command = "off" THEN
      DISPLAY "Light Switched OFF"

    ELSE IF command = "status" THEN
      DISPLAY "Light is Working"

    ELSE
      DISPLAY "Invalid Command"

    ENDIF

  END METHOD

END CLASS


CLASS SmartThermostat EXTENDS SmartDevice

  METHOD executeCommand(command)

    IF command = "on" THEN
      DISPLAY "Thermostat Activated"

    ELSE IF command = "off" THEN
      DISPLAY "Thermostat Deactivated"

    ELSE IF command = "status" THEN
      DISPLAY "Temperature = 25°C"
```

6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered


```
    ELSE
      DISPLAY "Invalid Command"

    ENDIF

  END METHOD

END CLASS


CLASS SmartLock EXTENDS SmartDevice

  PRIVATE PIN

  METHOD executeCommand(command)

    IF command = "on" THEN
      DISPLAY "Door Locked"

    ELSE IF command = "off" THEN
      DISPLAY "Door Unlocked"

    ELSE IF command = "status" THEN
      DISPLAY "Lock Status Displayed"

    ELSE
      DISPLAY "Invalid Command"

    ENDIF

  END METHOD
```

8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

```
    ELSE IF command = "status" THEN
      DISPLAY "Lock Status Displayed"

    ELSE
      DISPLAY "Invalid Command"

    ENDIF

  END METHOD

END CLASS


CREATE Light
CREATE Thermostat
CREATE Lock

STORE all devices in DeviceList

FOR EACH device IN DeviceList
   CALL device.executeCommand("status")
END FOR

END
```

Save Answer